FLAT PULLBACK PRESERVES RATIONAL EQUIVALENCE OF CYCLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a term by term pullback of a locally finite sum of divisors, moved inside the sum through the projection formula and reduced to the preparation formula on each fibre product

Editorial note. The label lemma-flat-pullback-rational-equivalence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Rational equivalence by locally finite sums of divisors of rational functions on integral closed subspaces of one dimension higher
- The preparation formula expressing the flat pullback of a divisor of a rational function as a multiplicity weighted sum of restrictions, and the compatibility of flat pullback with proper pushforward
- Local finiteness of the preimages of the subspaces occurring in a rational equivalence, which makes the pulled back sum well defined

Source reference. Original reference 02RG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1498–1498. Prerequisite pointer invoked at proof line(s) 2162; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02R9: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1198–1198. Prerequisite pointer invoked at proof line(s) 2154; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EPH: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2039–2039. Prerequisite pointer invoked at proof line(s) 2167; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RZ: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1991–1991. Prerequisite pointer invoked at proof line(s) 2172; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 2135; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let $\alpha \sim_{rat} \beta$ be rationally equivalent k-cycles on Y. Then $f^*\alpha \sim_{rat} f^*\beta$ as $(k+r)$-cycles on X.*

Proof. What do we have to show? Well, suppose we are given a collection

$$i_j : W_j \longrightarrow Y$$

of closed immersions, with each W_j integral of δ-dimension $k+1$ and rational functions $g_j \in R(W_j)^*$. Moreover, assume that the collection $\{|i_j|(|W_j|)\}_{j \in J}$ is locally finite in $|Y|$. Then we have to show that

$$f^*(\sum i_{j,*}\mathrm{div}(g_j)) = \sum f^* i_{j,*}\mathrm{div}(g_j)$$

is rationally equivalent to zero on X. The sum on the right makes sense by Lemma 02R9.

Consider the fibre products

$$i'_j : W'_j = W_j \times_Y X \longrightarrow X.$$

and denote $f_j : W'_j \to W_j$ the first projection. By Lemma 02RG we can write the sum above as

$$\sum i'_{j,*}(f_j^*\mathrm{div}(g_j))$$

By Lemma 0EPH we see that each $f_j^*\mathrm{div}(g_j)$ is rationally equivalent to zero on W'_j. Hence each $i'_{j,*}(f_j^*\mathrm{div}(g_j))$ is rationally equivalent to zero. Then the same is true for the displayed sum by the discussion in Remark 02RZ. □